Revive08 Station 1983 IFM Flow and Temperature Field Audit

Prepared by decarbX GHGsolve | June 26, 2026

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1. Executive Finding

The IFM SU6030 cumulative counters (`supply_total_l`, `return_total_l`) align with CAN bus fuel consumption at 100% agreement over a 4-hour engine-running window. The SkyView instantaneous `supply_flow_lph` and `return_flow_lph` fields are mis-scaled by approximately 59x relative to the cumulative-implied rate. The `fuel_temp_delta_c` field appears to compute `supply_temp_c − return_temp_c` instead of the correct `return_temp_c − supply_temp_c`. The sensors themselves are functional and transmitting valid data.

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2. Data Window

Parameter	Value
Asset	Revive08
Station	1983 (HRC REV POC1)
Pull file	`data/raw/station-1983-20260626-194441.csv`
Window	June 26, 2026 06:01Z to 18:04Z
Total rows	486
Channel values	28,188

Parameter	Value
Engine runtime used for validation	4.0 hours (12:16Z to 16:16Z)

3. CAN Bus Control Values

CAN bus is used as the trusted control source for fuel consumption validation.

Metric	Value
Total fuel consumed (`total_fuel_used_l` delta)	63.5 L
Average fuel rate during engine running	9.74 L/hr
Valid engine-running samples	449 rows

Runtime criteria applied: - `is_ignition_on` = 1 - `rpm` > 0 - `engine_fuel_rate_lph` > 0 - `engine_hours_min` increasing

4. IFM Cumulative Counter Validation

IFM flow was recalculated from cumulative counter deltas rather than the instantaneous `flow_lph` fields.

Metric	Value
`supply_total_l` delta (full window)	125.6 L
`return_total_l` delta (full window)	62.1 L
Net IFM fuel from cumulative counters	63.5 L
CAN bus total fuel	63.5 L
Percent agreement	**100.0%**

One-hour rolling window validation: 363 overlapping 1-hour windows were tested using `delta(supply_total_l) / delta(time)` and `delta(return_total_l) / delta(time)`. All 363 windows produced net IFM fuel rates matching CAN bus within rounding error.

Return/supply ratio: 44-51% depending on engine load. Consistent with diesel engine return system behavior (engine consumes 50-56% of supply, returns remainder).

5. Instantaneous flow_lph Problem

The `supply_flow_lph` and `return_flow_lph` fields report values that disagree with the cumulative counters from the same sensor hardware.

Field	Value
`supply_flow_lph` average (engine running)	0.53 L/hr
`return_flow_lph` average (engine running)	0.27 L/hr
Supply rate implied by `supply_total_l` cumulative delta	31.3 L/hr
Return rate implied by `return_total_l` cumulative delta	15.5 L/hr
Disagreement factor (supply)	**~59x**

The cumulative counters and the instantaneous rate fields are sourced from the same IFM SU6030 sensor hardware (`sensor_code: ifm_su6030`, `sensor_id: 6745` for supply; `sensor_id: 6746` for return). The cumulative counters are accurate. The instantaneous rate fields are not.

Recommendation: Do not use `supply_flow_lph`, `return_flow_lph`, or `net_fuel_flow_lph` until the conversion logic is corrected.

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6. Temperature Delta Issue

Raw temperature sensors read correctly. The derived delta field has the subtraction order reversed.

Field	Value
`supply_temp_c` average (engine running)	35.3°C
`return_temp_c` average (engine running)	44.9°C
Rows where return > supply	95.8% (430 of 449)
Physical delta: return - supply	+9.6°C (physically correct)
	-9.6°C

Field	Value
`fuel_temp_delta_c` average (SkyView derived)	
Apparent SkyView formula	`supply_temp_c − return_temp_c`
Correct formula should be	`return_temp_c − supply_temp_c`

The raw temperature sensors appear correctly placed on the engine supply and return fuel lines. A +9.6°C temperature rise from supply to return is physically consistent with diesel fuel passing through a hot engine. The `fuel_temp_delta_c` field reverses the sign.

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7. Sensor Status Question

Field	Value	Context
`supply_status`	0	All 486 rows
`return_status`	0	All 486 rows

Both status fields report `0` across the entire pull window. Data transmission continues uninterrupted despite status `0`. Please confirm the NAMUR NE 107 interpretation of `0` in the SkyView/ST910x implementation: - Does `0` indicate normal / data available? - Or does `0` indicate initialization, fault, maintenance required, or out-of-spec?

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8. Requested SkyView Actions

Priority	Action
P0	Review and correct `supply_flow_lph` and `return_flow_lph` conversion logic, units, scaling, and averaging window. The cumulative counter validation proves the sensor data path is intact. The instantaneous rate derivation has a ~59x discrepancy from the cumulative-implied rate.
P0	Correct `fuel_temp_delta_c` formula to `return_temp_c − supply_temp_c`.

Priority	Action
	Currently appears to compute `supply_temp_c − return_temp_c`, which reverses the sign of a physically correct measurement.
P1	Confirm the NAMUR NE 107 meaning of sensor status value `0` for IFM SU6030 channels.
P2	Re-run `fuel_flow_quality_low_ifm_flow` and `fuel_flow_quality_return_gt_supply` flags after `flow_lph` conversion is corrected. Current flags are likely false positives driven by the mis-scaled rate fields.

9. Attachments / References

- **CSV data:** `data/raw/station-1983-20260626-194441.csv` (486 rows, 58 columns)
- **JSONL data:** `data/raw/station-1983-20260626-194443.jsonl` (486 records, 47 channels per record)
- **Channel metadata:** `data/raw/station-1983-20260626-194441.csv.channels.json` (57 registered channels)
- **Validation script:** Python cumulative-delta recalculation, available on request
- **Previous reports:** `IFM_VALIDATION_CHECKLIST.md`, `STATION_1983_VALIDATION_REPORT.md` (updated June 26, 2026)

End of Report